

Emanuele Bugliarello

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Research Interests

Multi-modal and multilingual natural language understanding, representation learning, transfer learning

Education

PhD in Computer Science 2019–present
University of Copenhagen Copenhagen, Denmark
Advisors: Desmond Elliott, Anders Søgaard

MSc in Communication Systems, Data Analytics specialization 2015–2018
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)

BEng in Electronics and Information Engineering 2013–2018
Tongji University (同济大学) Shanghai, China
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30

BSc in Telecommunications Engineering 2012–2015
Politecnico di Torino Turin, Italy
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 Cum Laude

Experience

Research Associate with Prof. Naoaki Okazaki 10/2018–09/2019
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学) Tokyo, Japan
On Neural Machine Translation

- Design of an effective syntactically-aware self-attention mechanism for the Transformer model

Visiting Graduate Student with Prof. Rui Fan and Prof. Dan Alistarh 04/2018–07/2018
ShanghaiTech University (上海科技大学) Shanghai, China
Large-Batch Algorithms for Deep Learning at Scale

- Implementation of state-of-the-art algorithms for large-batch training of Convolutional Neural Networks
- Evaluation of large-batch training algorithms performance on different datasets to assess their generality

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot 09/2017–03/2018
Technicolor AI Lab Palo Alto, CA, USA
Intelligent VFX Scheduling

- Predicting visual effects (VFX) assets overspending by blending classification and censored regression models
- Predicting weekly number of required employees in VFX departments using advanced regression methods
- Recommending employees to VFX departments with novel structured matrix factorization models

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller Summer 2017
Swiss National Supercomputing Centre (CSCS) Lugano, Switzerland
Distributed TensorFlow: A performance evaluation

- Deployment of distributed TensorFlow on the Piz Daint supercomputer and various AWS EC2 clusters
- Scalability study of TensorFlow using InceptionV3 for image classification on ImageNet

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308)

Spring 2017

Publications

E. Bugliarello, N. Okazaki. Improving Neural Machine Translation with Parent-Scaled Self-Attention. arXiv preprint 2019.

E. Bugliarello, S. Jain, V. Rakesh. Matrix Completion in the Unit Hypercube via Structured Matrix Factorization. IJCAI 2019.

Projects

Evaluating the evolution of Wikipedia's navigability

Spring 2017

Semester project with Prof. Robert West

Data Science Lab @ EPFL

Investigate Wikipedia's navigability over the course of its history, based on the English Wikipedia's complete edit history

- Design of a parser to extract hyperlinks (including hatnotes) from Wiki Markup Language
- Processing of a big data set (6 TB) consisting of all the revisions in the English Wikipedia with Spark
- Analysis of the temporal evolution of Wikipedia's graph properties
- Poster accepted at Wiki Workshop, WWW 2018

Solarity: Solar Clarity for Swiss Households

Fall 2016

Applied Data Analysis (CS-401)

EPFL

Based on historical weather data, predict solar energy production in the coming years to provide information on initial investment, break-even time, and potential savings for domestic solar installations

- Development of a Web scraper to retrieve publicly available weather data
- Predicting solar potential in the coming years with a regression model based on historic data
- 2nd best class project award (over 60+ projects)

Twitter Sentiment Classification

Fall 2016

Machine Learning (CS-433)

EPFL

Examine the accuracy of different learning algorithms in determining sentiments associated to tweets

- Text preprocessing: replace repeating letters, stemming, n-grams, remove stop words
- Word embeddings: TF-IDF, GloVe
- Classifiers: Logistic Regression, multinomial naive Bayes, SVM with linear kernel

Skills

Programming

- Python, C, Java, Matlab, Java for Android
- NumPy, Pandas, scikit-learn, CUDA
- PyTorch, TensorFlow
- Spark, Hadoop MapReduce, MPI
- SQL, MySQL, Mongo

Languages

- Italian (native)
- English (fluent – TOEFL iBT 109)
- French (intermediate)
- Chinese (intermediate – HSK III)
- Japanese (basic)

Fellowships, Honors, and Awards

University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship

06/2019

EPFL PhD Program in Computer Science Fellowship [offer declined]

05/2019

The Japanese Government (MEXT) Scholarship [offer declined]

01/2019

EPFL TCP/IP 2015-2016 Research Exercise Award

12/2015

PoliTong Merit Scholarship

07/2013

MIUR National Register of Excellences

06/2012